



STRENGTHENING HEALTH SYSTEMS AND RMNCAH-N OUTCOMES THROUGH RAPID CYCLE ANALYTICS AND DATA USE

Country Workshop: Introduction to FASTR RMNCAH-N Service Use Monitoring

Lusaka, Zambia

January 27-30, 2026

Welcome and Opening Remarks



Introductions



Workshop Objectives

The objectives of the workshop are to:

- **Introduce the FASTR HMIS analysis platform** – Familiarize participants with the platform's purpose, core features, and functionalities, including workflows, reporting options, and outputs
- **Strengthen analytical and reporting skills** – Enhance participants' capacity to manage data structures, customize visualizations, and produce tailored reports for diverse audiences
- **Prepare MoH for participation in the multi-country workshop** – Configure the FASTR analytics platform for Zambia; Produce the first quarterly report
- **Adapt the FASTR phone survey questionnaire to the Zambian context**

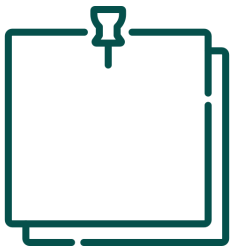
Expectations for the Workshop

Write on a sticky note and stick it on the flip chart:

What do you want to learn?

What questions about FASTR do you want answered?

We will review these on day four to see if we met our expectations.



Expected Outputs

Disruption Analysis Activities

- Data extraction and configuration of analytics platform
- Produce first quarterly report
- Review results and contextualize findings (with relevant program teams)
- Roadmap for participation in Abuja workshop

Phone Survey Activities

- Adaptation of questionnaire (with program input)

Agenda

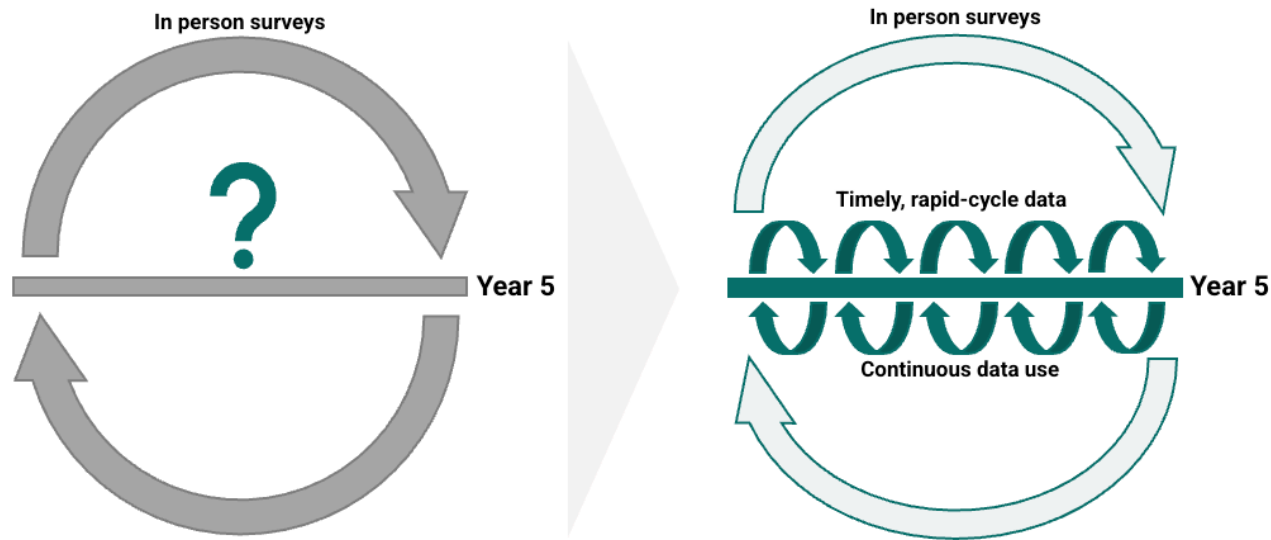
Day 1 -- Laying the Foundation: Introducing FASTR and Configuring the Analytics Platform

Time	Agenda	Facilitator/Presenter
Opening Session		
08:30–09:00	Participant registration	MoH team
09:00–09:10	Welcome and opening remarks	MoH team
09:10–09:20	Icebreakers/Introductions	MoH team
09:20–09:35	Workshop objectives and agenda	GFF FASTR team
Session 1: Overview of the FASTR approach		
09:35–10:15	Overview: FASTR Approaches	GFF FASTR team
Session 2: Identifying priority questions and indicators		
10:15–11:00	Identifying priority questions and indicators	GFF FASTR team
Session 3: HMIS data extraction		
11:00–12:00	Data extraction: Rationale and methods	GFF FASTR team
Session 4: Getting into the platform		
12:00–12:30	Getting participants into the platform	GFF FASTR team
12:30–14:00	<i>Lunch Break</i>	
Session 5: Configuring the FASTR analytics platform		
14:00–15:30	Configuring the analysis platform	GFF FASTR team
Closing		
15:30–16:00	Key messages and wrap-up	GFF FASTR team
16:00–16:30	Reflections from participants	Participants

Session 1: Overview of the FASTR approach

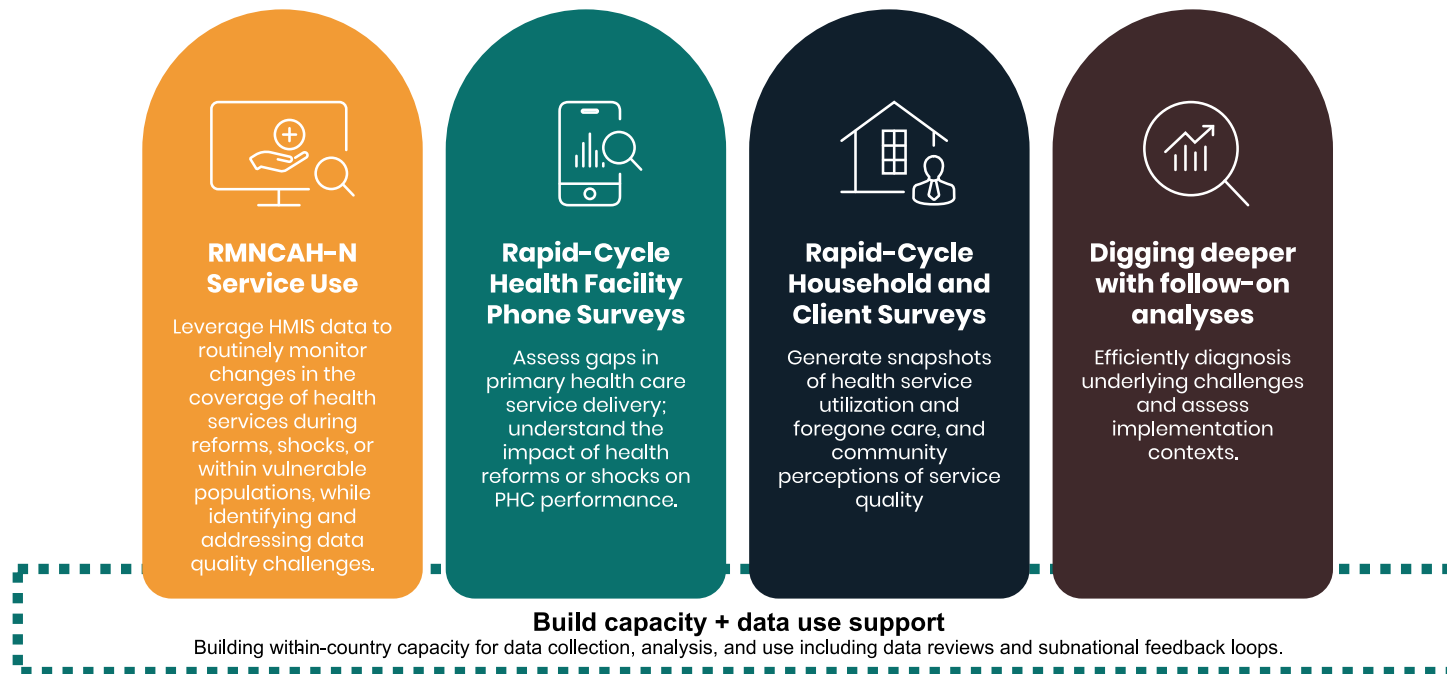
What are we trying to achieve?

Rapid cycle analytics accelerates improvements in RMNCAH-N outcomes by increasing the systematic use of data for decision making



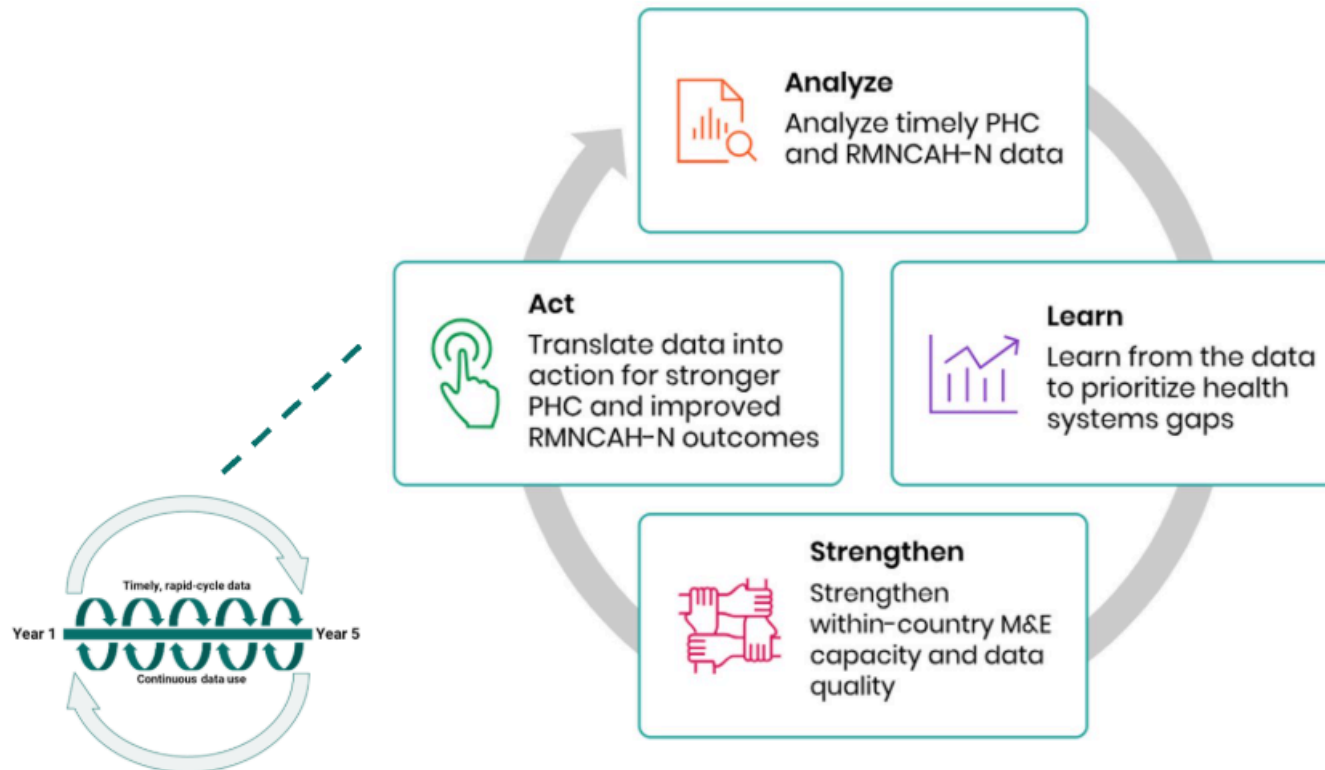
How can this be achieved?

Timely, rigorous, and low-cost approaches to monitoring PHC systems, underpinned by capacity building and data use support aligned with country demand and needs



What is FASTR?

An approach to catalyzing continuous 'analyze, learn, strengthen, act' cycles to drive the systematic use of timely data for decision making.

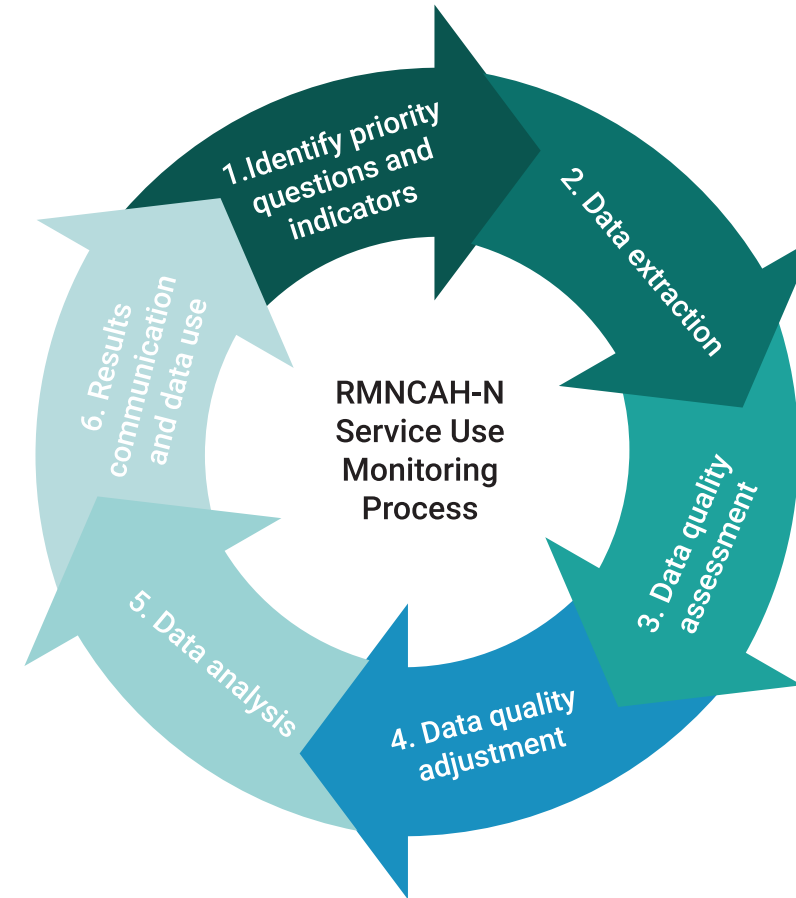


What is the FASTR approach to RMNCAH-N service use monitoring?

Quarterly analyses of DHIS2 data, focusing on prioritized national indicators

Building sustainable tools to ensure that stakeholders who need to use data can generate the right analysis and visualizations, at the right time, on their indicators of interest

Combining analysis and visualization with capacity strengthening and data use support for sustainability and institutionalization



Demo – FASTR analysis platform



Session 2: Identifying priority questions and indicators

What is a data use case?

A data use case is a specific scenario where data is utilized to achieve a particular goal or solve a problem.

Why is defining a data use case important?

- Guides decision making by providing a clear framework for analysis
- Enhances efficiency by focusing analyses on a set of relevant key indicators to solve a specific data need
- Leads to better results by aligning data efforts with organizational goals

Data use case: Application 1

Placeholder – content to be added

Data use case: Application 2

Placeholder – content to be added

Data use case: Application 3

Placeholder – content to be added

What is our common data use case that will be the focus of this workshop?

Following large shifts in resource availability from external sources, many countries are experiencing abrupt and dramatic reductions in financing

- Resulting in critical gaps in programs and systems
- Leading to potentially severe effects on service delivery and health outcomes for women, children and adolescents

Key questions arising:

- What is the magnitude of the cuts, and what effect are they having on service delivery?
- What is the optimal way to prioritize remaining resources?
- What other adaptations can safeguard and strengthen essential service delivery for women, children and adolescents?

How do we select indicators? What makes a good indicator for the FASTR analysis?

Indicator selection is critical to the quality and usefulness of FASTR analysis. Indicators should be chosen based on the following criteria:

- **Relevance** – Does this indicator provide data that aligns with our priority questions and objectives?
- **Volume** – Is this indicator collected at a high volume, which improves the robustness of analysis?
- **Completeness** – Does the indicator have a high completeness rate across reporting facilities?
- **Frequency** – Is the indicator reported frequently enough (e.g., monthly) to support rapid-cycle analysis?
- **Type** – Is this indicator a count of services delivered?

FASTR core indicators

The FASTR approach focuses on a core set of RMNCAH-N indicators that represent key points along the reproductive, maternal, newborn, child, and adolescent health and nutrition continuum in low- and middle-income countries.

These indicators typically have higher reporting volumes and completeness and serve as proxies for broader service delivery patterns.

- Antenatal client 1st visit
- Antenatal client 4th visit
- Institutional delivery
- Postnatal care 1
- BCG doses
- Pentavalent 1st dose
- Pentavalent 3rd dose
- Outpatient visits

Why focus on high-volume indicators?

One of the core strengths of the FASTR approach is its ability to adjust for data quality issues. High-volume indicators are better suited to this process because:

Reduced sensitivity to outliers

In low-volume indicators, individual data points can disproportionately affect trends.

More stable estimates

High-volume data reduce random variability and improve the reliability of trend detection.

Clearer identification of true anomalies

Larger counts make it easier to distinguish genuine outliers from natural variation.

Why focus on high-completeness indicators?

Indicators with high reporting completeness are preferred because they:

Improve data reliability

More complete data reduces bias and provide a more representative picture of service delivery.

Support consistent analysis

High completeness enables meaningful comparisons across time and geographic areas.

Reduce misinterpretation

Incomplete data can falsely suggest changes in service utilization when changes are driven by reporting gaps rather than real trends.

While statistical methods such as imputation can be used to address incomplete data, these methods require assumptions about missing values.

Why focus on count indicators?

Limitations of proportion indicators

- Proportions limit the ability to adjust numerators and denominators separately for data quality issues
- Numerators and denominators may each be affected by different sources of error
- Separating counts from denominator estimation allows for more transparent and flexible adjustment

Mortality as a rare event

- Mortality indicators are typically low-frequency and not well suited to adjustment
- These indicators are generally better analyzed using annual rather than monthly or quarterly data

Countries have selected indicators to align with the disruptions context and country priorities

		Indicator mapping for RMNCAH-N Service Use Monitoring using Administrative Data				
Data Source Mapping				Indicator Definition		
INDICATOR CATEGORY	INDICATOR OF INTEREST	INDICATOR SOURCE (national strategy, WB project, FASTR core indicators, etc)	IS THIS INDICATOR IN DHIS2?	OFFICIAL INDICATOR NAME IN DHIS2	INDICATOR JSON ID	NOTES
RMNCAH+N	ANC1	FASTR core indicators				
RMNCAH+N	ANC4	FASTR core indicators				
RMNCAH+N	Deliveries - institutional	FASTR core indicators				
RMNCAH+N	BCG	FASTR core indicators				
RMNCAH+N	PNC1	FASTR core indicators				
RMNCAH+N	Penta1	FASTR core indicators				
RMNCAH+N	Penta3	FASTR core indicators				
RMNCAH+N	OPD visits	FASTR core indicators				

- The FASTR Data Prep Checklist has been shared with countries
- The checklist includes the FASTR core RMNCAH-N indicators
- Countries have added additional indicators relevant to their context (e.g., services impacted by funding shifts, government priorities)
- These are the indicators included in the current analysis
- Countries can continue to add indicators over time as needed

Activity



We are going to review the Zambia data prep checklist together and discuss for each indicator:

Relevance

Does this indicator provide data that aligns with our priority questions and objectives?

Volume

Is this indicator collected at a high volume, which improves the robustness of analysis?

Completeness

Does the indicator have a high completeness rate across reporting facilities?

Frequency

Is the indicator reported frequently enough (e.g., monthly) to support rapid-cycle analysis?

Type

Is this indicator a count of services delivered?

Based on the discussion we can refine the selection of indicators.

Session 3: HMIS data extraction

Show of hands...



Do you regularly extract data from DHIS2?

If so, what are the primary reasons?

Why would you extract data from DHIS2? Why not just do analysis in DHIS2 itself?

Data quality adjustment

The FASTR approach focuses on data quality adjustments to expand the analyses countries can do with DHIS2 data and to generate more robust estimates.

Analysis complexity

The FASTR approach uses more advanced statistical methods, such as regression analysis, which are not available in DHIS2. While DHIS2 can plot trends over time using raw data, FASTR can go further by identifying significant increases or decreases in service volume, adjusting for data quality issues, accounting for expected seasonal variations, and comparing key periods, such as before and after a reform.

The choice between DHIS2 and the FASTR approach should be guided by the specific purpose of your analysis. Select the tool that best aligns with your analytical needs!

Data format and granularity

Data format example

Data should be downloaded for each indicator of interest, at facility level, and monthly for the period of interest.

Data should be saved in long format meaning each row represents a single observation or measurement (see example).

Data should be saved in .csv format and can be saved in either a single .csv file or multiple .csv files which will be combined when uploading to the analysis platform.

How much data?

Initial FASTR analysis

- Generally recommended to download approximately five years of historical data
- However, the exact period should be determined based on data availability, consistency in indicator definitions over time, and the specifics of a country's routine data system
- Ideally, using at least five years of historical data allows for a thorough assessment of trends over time

Routine update to FASTR analysis

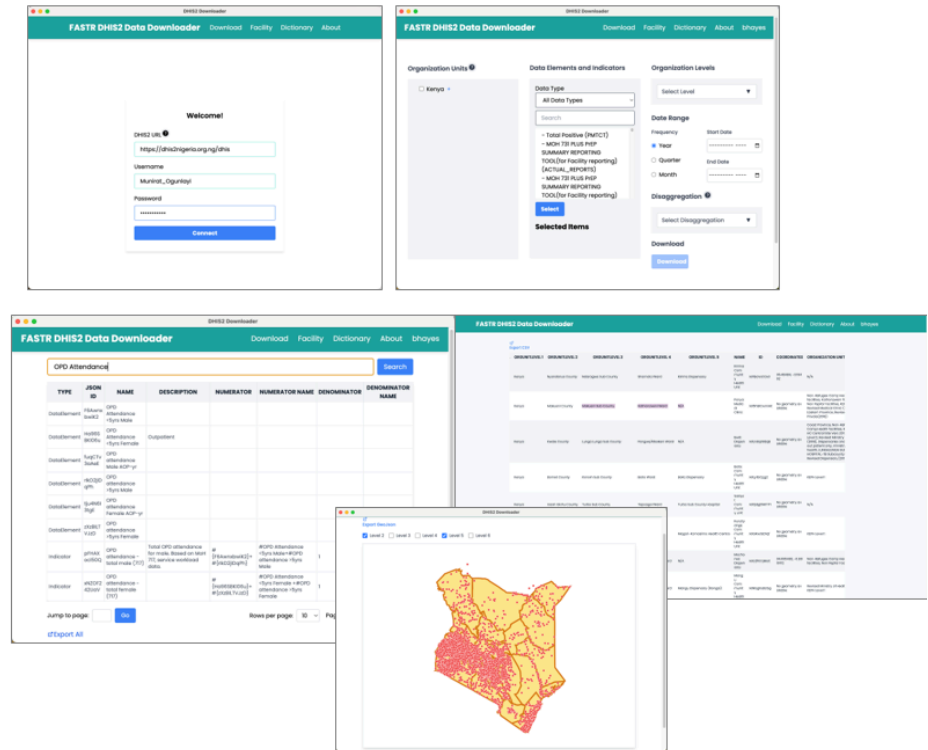
- Start with the existing database and download new data covering the most recent months not previously included – this is usually a three-month period when the FASTR analysis is being implemented on a quarterly basis
- Additionally, include the three preceding months to the new data time period, as this relatively recent data is often subject to changes due to late reporting or data quality adjustments
- If you have reason to believe there have been substantial changes to the historical data, you can always choose to redownload a longer time period

Data extraction

We offer two tools for bulk DHIS2 data extraction: a user-friendly Data Downloader and a direct import feature within the FASTR analytics platform.

The Data Downloader provides a streamlined interface to download DHIS2 data. This tool is particularly useful to explore DHIS2 metadata and download indicators requiring disaggregated dimensions.

The Data Downloader is available at:
<https://github.com/worldbank/DHIS2-Downloader/releases/>



Activity: Data downloader – LIVE Demo and practice



Screenshot of the DHIS2 Data Downloader interface:

The interface is titled "DHIS2 Downloader" in the browser window. The main header is "FASTR DHIS2 Data Downloader" with navigation links: "Download", "Facility", "Dictionary", "About", and language toggles "EN" and "FR".

The central form is titled "Welcome!" and contains the following fields:

- DHIS2 URL**: A text input field with a help icon.
- Username**: A text input field.
- Password**: A password input field with a toggle icon.
- Connect**: A blue button to submit the form.

Session 4: Getting into the platform

FASTR analytics platform

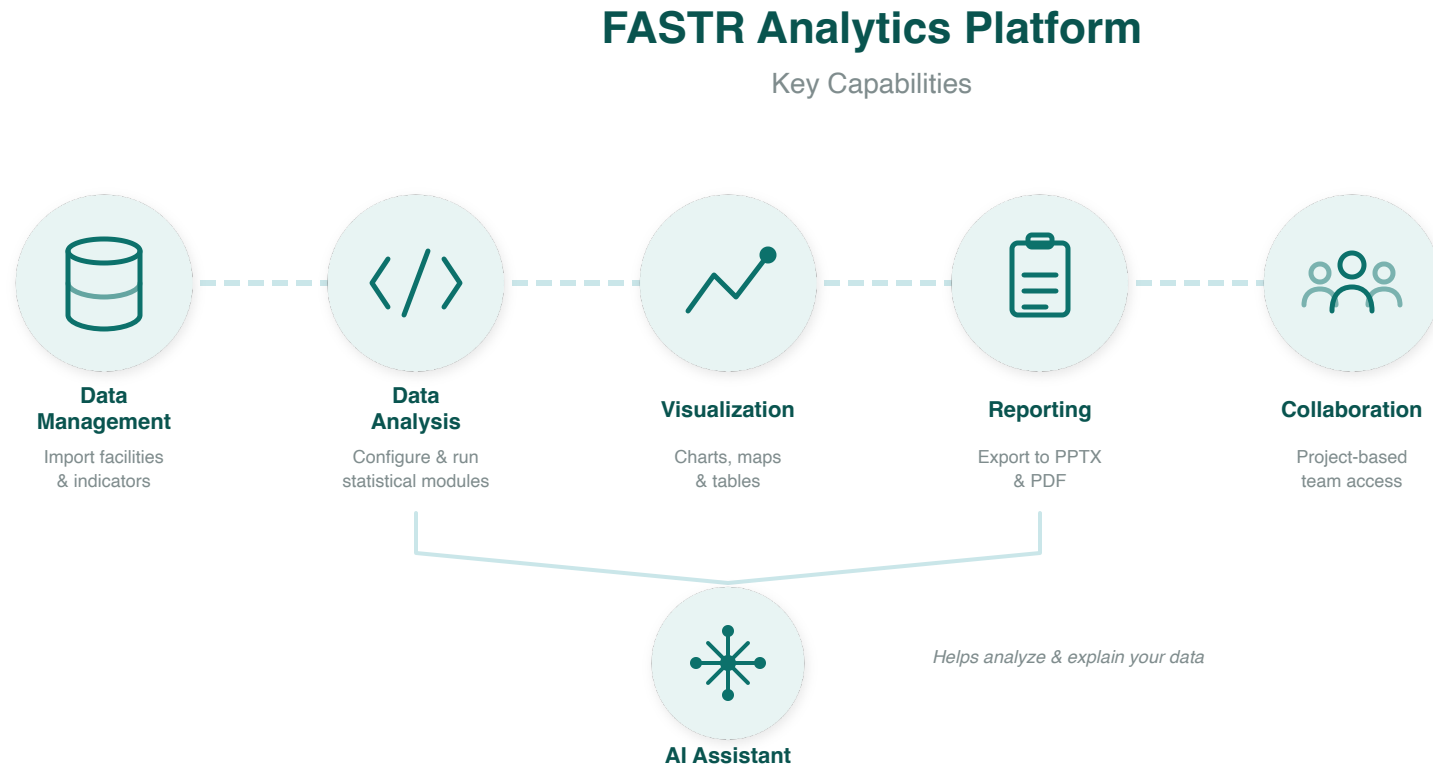
The FASTR analytics platform is a web-based tool designed to support data quality assessment, adjustment, and analysis for routine health data.

It allows users to upload and analyze data from various sources, including DHIS2, with built-in statistical methods to generate an adjusted dataset and run priority analyses on selected indicators.

The platform provides a user-friendly interface for running analyses and offers flexible options for visualizing and exporting results.

 FASTR platform

Platform Capabilities



Data flows from import through analysis to shareable outputs.

Country Instance

Each country has its own **instance** of the FASTR analytics platform.

An instance contains:

- All registered users and their accounts
- The shared administrative structure (regions, districts, facilities)
- Indicator definitions and data sources
- All projects created for that country

Think of an instance as your country's dedicated workspace.

User Roles and Permissions

There are two levels of permissions in the platform:

Instance-level roles:

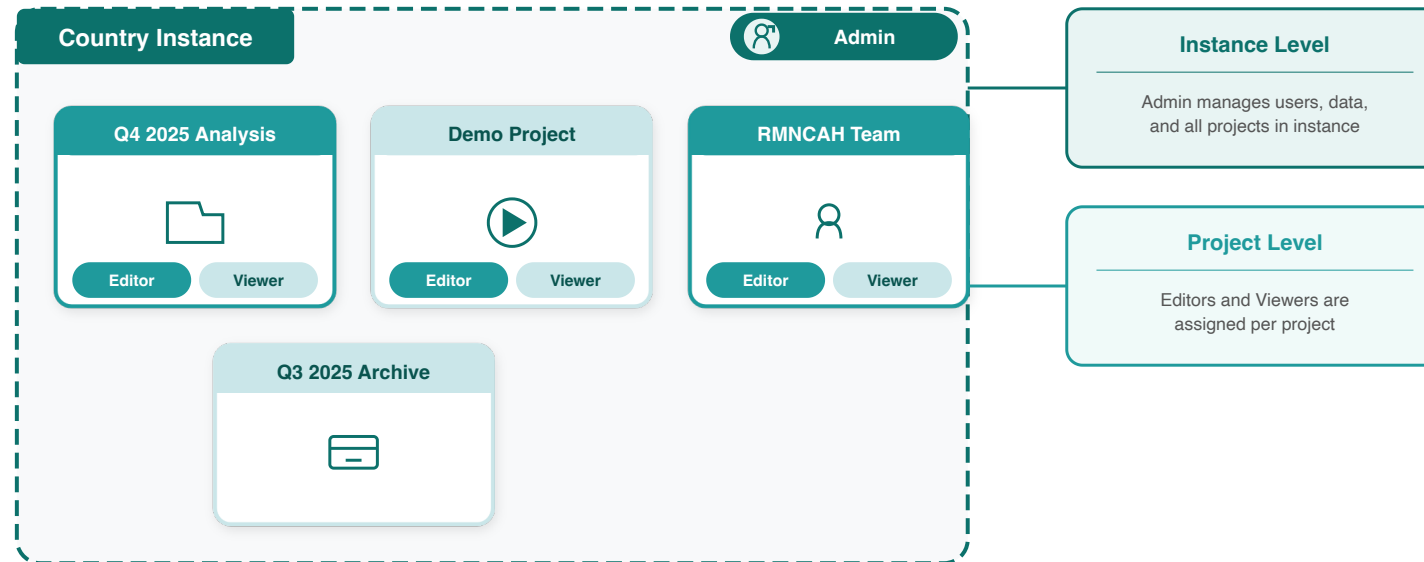
- **Instance Administrators** can add users, create projects, assign roles, upload data, import and configure modules, and run analyses

Project-level roles:

- **Project Editors** can create visualizations, create reports, and download/export results
- **Project Viewers** can view visualizations, view reports, and download/export results

Administrators are assigned per instance; Editors and Viewers are assigned per project.

Projects Within an Instance



Each country instance can contain **multiple projects**.

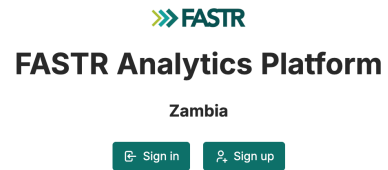
A country may only need one project, or multiple projects can be used for:

- Different versions of analyses
- A demo or playground project
- Separate projects for different teams or programs

Key questions when setting up:

- Who is the admin?
- Who can edit?
- Who can view?

Practice: Logging Into the Platform

The image shows a "Create your account" form. It includes a welcome message, fields for "First name" and "Last name", an "Email address" field, and a "Password" field with a toggle for visibility. A "Continue" button is at the bottom, and a link to "Sign in" is at the very bottom for users who already have an account.

1. Go to <https://zambia.fastr-analytics.org>

3. Enter your information (verify email)

2. Click Sign up and enter your details

4. After login, you'll be added to a project



Lunch Break

90 minutes

Back at 14:00

Session 5: Configuring the FASTR analytics platform

Configuring the analysis platform

- Configuration of the analysis platform is an admin feature
- We will work together to configure the following items:
 - Admin areas (regions, districts)
 - Facility structure
 - Indicator definitions
- Note since this is an admin feature all participants will NOT be doing this step. Instead, you will select one person to have admin rights, and they will help us walk through these steps.

Activity: Setting Up Admin Areas



In this hands-on session, we will configure:

- Admin areas (regions, districts)
- Facility structure
- Indicator definitions

Participants will work directly in the platform

Activity: Importing Data



In this hands-on session, we will:

- Review data format requirements
- Walk through the import process
- Handle validation and error checking

Participants will import their country's data

Activity: Installing and Running Modules



In this hands-on session, we will:

- Review available analysis modules
- Install required modules
- Run initial analyses

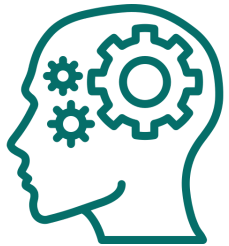
Participants will configure and run modules on their data

Key messages and wrap-up

Reflections from participants

Day 2

Recap of day 1



Agenda

Day 2 -- Building the Analysis: Applying FASTR Methods and Generating Outputs

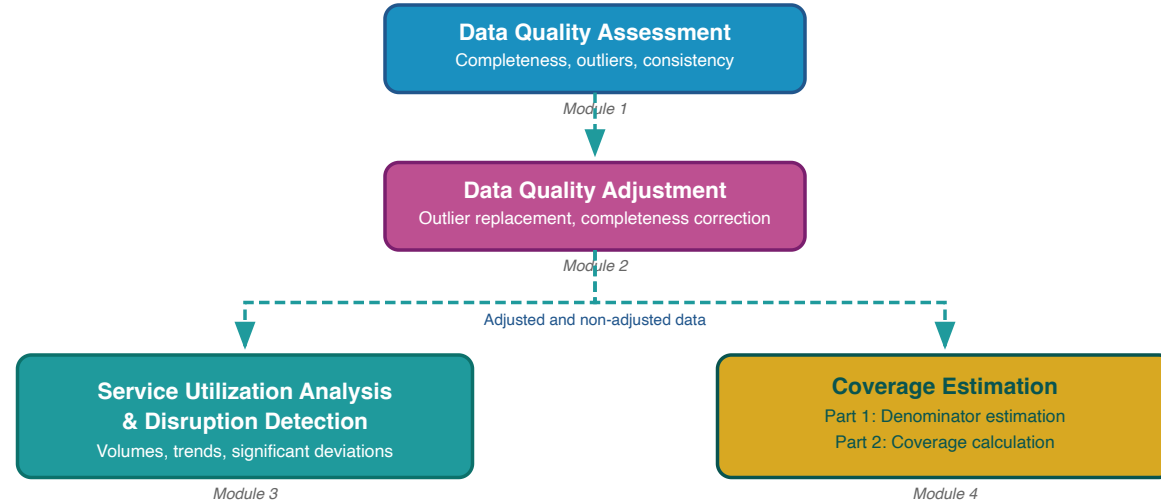
Time	Agenda	Facilitator/Presenter
09:00–09:15	Recap and Day 2 agenda	GFF FASTR team
Session 6: Overview of FASTR methods and interpretation		
09:15–10:15	Data quality, service utilization, coverage: Methods and interpretation	GFF FASTR team
Session 7: Creating a project		
10:15–11:15	Project creation and settings	GFF FASTR team
Session 8: Creating visualizations		
11:15–12:30	Creating and editing visualizations	GFF FASTR team
12:30–14:00	<i>Lunch Break</i>	
Session 9: Creating reports		
14:30–16:30	Practice creating and editing reports	GFF FASTR team

Session 6: Overview of FASTR methods and interpretation

FASTR analytical pipeline

The FASTR analysis follows a sequential workflow:

- 1. Assess data quality** – Identify issues with completeness, outliers, and consistency
- 2. Adjust for quality issues** – Apply corrections to improve data reliability
- 3. Analyze adjusted data** – Generate service utilization and coverage estimates



FASTR approach to data quality

FASTR takes a multi-pronged approach, based on the belief that **data quality should not be a barrier to data use**.

- Conduct granular, facility-level data quality assessments
- Focus on high-volume indicators that produce more stable estimates
- Emphasize variation over time and space rather than point estimates
- Interpret results collaboratively with in-country decision-makers

Using data and providing feedback is viewed as the first step toward improving data quality.

Rationale for data quality assessment

Challenge: Routine health facility data may contain quality limitations:

- Reported values outside plausible ranges
- Reporting gaps affecting completeness
- Inconsistencies between related indicators

Implications: These limitations can distort decision-making:

- Inaccurate assessment of service delivery trends
- Misidentification of areas needing intervention
- Suboptimal resource allocation

Objectives of data quality assessment

Objective 1: Enable analytical adjustment

Systematic assessment supports targeted adjustments, enhancing the utility of HMIS data for evidence-based decision-making.

Objective 2: Monitor data quality trends

- Inform indicator selection based on quality profiles
- Guide targeted interventions and supportive supervision
- Evaluate effectiveness of improvement initiatives over time

Indicator completeness

Indicator completeness measures whether facilities that should be reporting data on specific indicators are actually doing so. This is different from overall reporting completeness – we're looking at specific data elements, not just whether the monthly form was submitted.

Definition: Percentage of facilities reporting each month out of facilities expected to report.

- A facility is "reporting" if there is a non-missing, non-zero value for the indicator that month
- A facility is "expected to report" if it has reported any volume for that indicator within the past year

Higher and stable completeness improves data reliability.

Indicator completeness output

Indicator Completeness

Percentage of facility-months with complete data, Jan 2022 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
District 001	93.9%	90.3%	83.3%	70.8%	82.8%	92.3%	91.9%	90.4%	94.9%
District 002	88.5%	85.9%	82.0%	64.2%	79.8%	90.8%	88.6%	88.1%	88.4%
District 003	89.8%	87.0%	85.4%	53.4%	78.5%	86.2%	86.3%	85.5%	88.3%
District 004	90.7%	82.6%	84.5%	67.6%	82.1%	89.1%	90.4%	89.7%	90.4%
District 005	89.7%	84.0%	81.3%	60.4%	80.9%	84.3%	84.2%	83.6%	89.5%
District 006	85.5%	79.4%	74.5%	56.8%	74.5%	85.3%	82.9%	79.2%	85.6%
District 007	97.2%	93.8%	91.8%	73.5%	90.8%	91.3%	95.2%	87.9%	97.0%
District 008	93.1%	87.6%	85.5%	44.6%	71.8%	79.9%	87.2%	87.5%	95.0%
District 009	95.7%	75.8%	85.0%	34.9%	77.6%	83.5%	92.2%	78.1%	95.9%
District 010	99.2%	99.6%	95.8%	98.4%	94.2%	94.6%	93.2%	88.7%	100.0%
District 011	98.0%	95.2%	94.4%	66.6%	91.7%	96.1%	95.4%	91.8%	98.2%
District 012	92.6%	86.4%	83.7%	64.5%	84.7%	84.7%	87.2%	86.7%	92.6%
District 013	93.4%	90.8%	90.7%	77.1%	88.2%	90.3%	90.8%	87.7%	93.1%
District 014	88.8%	79.1%	81.2%	77.9%	81.2%	90.0%	89.8%	86.3%	91.2%
District 015	93.0%	88.2%	85.3%	63.9%	83.1%	86.7%	88.8%	88.7%	93.1%
District 016	96.0%	85.1%	92.0%	65.7%	91.6%	96.1%	95.5%	88.6%	96.8%
District 017	90.1%	84.8%	90.4%	53.4%	83.0%	72.7%	79.3%	76.0%	89.5%
District 018	97.1%	95.9%	93.2%	79.8%	92.3%	95.4%	95.8%	92.9%	97.9%

90% or above

80% to 89%

Below 80%

Higher completeness improves the reliability of the data, especially when completeness is stable over time. Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. A high completeness does not indicate that the HMIS is representative of all service delivery in the country, as some services may not be delivered in facilities, or some facilities may not report.

What you see: Heatmap showing completeness by indicator and region over time.

Formula: Completeness % = (facilities reporting / facilities expected) × 100

Interpretation: Look for systematic gaps by region or indicator, declining trends, or seasonal patterns. Low completeness suggests reporting barriers needing attention.

Outlier detection

Outliers are values that are suspiciously **high** compared to a facility's usual reporting volume. They may result from data entry errors or genuine programmatic changes (e.g., campaigns).

Note: FASTR only flags high values as outliers – unusually low values are not flagged, as these more likely reflect service disruptions than data errors.

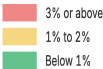
How outliers are identified: For each facility and indicator, we assess within-facility variation in monthly reporting. A value is flagged if it deviates significantly from the facility's typical pattern (using statistical thresholds based on median absolute deviation).

Outlier detection output

Outliers

Percentage of facility-months that are outliers, May 2024 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
Region 001	0.6%	0.2%	0.2%	1.9%	1.2%	0.6%	0.9%	0.6%	0.9%
Region 002	0.6%	0.0%	1.2%	1.0%	1.8%	0.2%	0.2%	0.1%	2.7%
Region 003	0.5%	0.4%	0.1%	0.0%	0.1%	0.1%	0.4%	0.6%	1.4%
Region 004	0.4%	0.0%	0.0%	0.8%	0.0%	0.5%	0.8%	0.9%	0.1%
Region 005	0.5%	0.9%	0.5%	0.8%	0.5%	0.6%	0.8%	0.3%	3.3%
Region 006	0.4%	0.0%	0.1%	0.3%	0.1%	1.3%	2.3%	2.5%	0.6%



Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100.

What you see: Heatmap showing proportion of values flagged as outliers by indicator and region.

Formula: $\text{Outlier \%} = (\text{values flagged} / \text{total values}) \times 100$

Interpretation: High rates may indicate data entry errors or legitimate events like campaigns. Review facility registers to distinguish between the two.

Internal consistency

Internal consistency checks whether related indicators maintain expected logical relationships. For example, ANC1 should always be greater than or equal to ANC4 (you can't have a 4th visit without a 1st visit). Similarly, Penta1 $\geq$ Penta3.

Why it matters: When these relationships are violated, it signals potential data quality issues – either double-counting, under-reporting, or errors in how data flows through the system.

What FASTR checks: For each geographic area, the module assesses whether indicator pairs maintain expected ratios across the reporting period.

Internal consistency output

Internal consistency

Percentage of sub-national areas meeting consistency benchmarks, May 2024 to Apr 2025

	ANC1 is larger than ANC4	Delivery is approximately equal to BCG	Penta1 is larger than Penta3
Region 001	100.0%	0.0%	97.2%
Region 002	100.0%	0.0%	100.0%
Region 003	100.0%	45.8%	83.3%
Region 004	100.0%	37.5%	87.5%
Region 005	100.0%	0.0%	87.5%
Region 006	100.0%	49.0%	84.4%

- 90% or above
- 80% to 89%
- Below 80%

Internal consistency assesses the plausibility of reported data based on related indicators. Consistency metrics are approximate - depending on timing and seasonality, indicator definitions, and the nature of service delivery and reporting, values may be expected to sit outside plausible ranges. Indicators which are similar are expected to have roughly the same volume over the year (within a 30% margin). The data in this analysis is adjusted for outliers.

What you see: Heatmap showing % of areas where indicator pairs meet expected relationships.

Formula: Consistency % = (areas meeting criteria / total areas) × 100

Interpretation: Low consistency may indicate data flow problems, double-counting, or systematic under-reporting.

Overall data quality score

The overall DQA score combines all three quality dimensions – completeness, outliers, and consistency – into a single measure. For each indicator in each time period, it calculates what proportion of monthly values pass ALL quality criteria.

Why it matters: This provides a summary view to quickly identify which indicators and areas have the most data quality challenges, helping prioritize improvement efforts.

Overall data quality score output

Overall DQA score

Percentage of facility-months with adequate data quality over time

	2022	2023	2024	2025
Region 001	60.8%	76.6%	76.4%	84.4%
Region 002	55.3%	61.2%	58.2%	52.0%
Region 003	69.3%	73.4%	60.6%	47.0%
Region 004	54.6%	70.7%	70.0%	84.9%
Region 005	57.9%	69.5%	56.6%	55.4%
Region 006	74.0%	84.7%	72.2%	71.7%

- 80% or above
- 70% to 79%
- Below 70%

Adequate data quality is defined as: 1) No missing data or outliers for OPD, Penta1, and ANC1, where available 2) Consistent reporting between Penta1/Penta3 and ANC1/ANC4.

What you see: Heatmap showing overall data quality by indicator and region.

Formula: % adequate = (values meeting ALL criteria / total values) × 100

Interpretation: 100% = all values pass all checks. Use this to prioritize DQ improvement and select indicators for analysis.

Data quality adjustment

Why adjust? Outliers and reporting gaps identified in the DQA will distort service utilization and coverage estimates if left uncorrected. The goal is to replace problematic values with reasonable estimates based on each facility's own historical patterns.

How? Outliers and missing values are replaced using 6-month rolling averages from the facility's historical data.

Four parallel datasets: FASTR produces unadjusted, outliers-only adjusted, completeness-only adjusted, and both-adjusted versions. This enables sensitivity analysis – comparing results across scenarios to assess how much conclusions depend on adjustment choices.

Excluded from adjustment: Mortality indicators (discrete events that shouldn't be smoothed) and low-volume indicators (<100 events/month, where adjustment adds noise).

Outlier adjustment output

Deviance Due to Outliers

Percent change in volume due to outlier adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods- long acting	Family planning methods- short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	0.0%	0.2%	4.6%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	0.2%
Bombali District	1.3%	0.0%	0.0%	2.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%
Bonthe District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.5%	0.0%	0.0%	0.0%
Falaba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.7%	0.0%	0.0%	2.2%	0.0%	0.0%
Kailahun District	0.0%	0.0%	0.6%	0.0%	0.0%	0.0%	0.0%	0.0%	4.5%	2.1%	0.2%	0.2%
Kambia District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Karene District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.7%	1.3%	5.7%	4.7%	0.0%	0.0%
Kenema District	0.0%	0.0%	0.0%	0.0%	0.0%	1.2%	1.7%	1.4%	1.8%	0.4%	0.0%	0.0%
Koinadugu District	0.0%	0.0%	0.0%	0.0%	0.3%	0.0%	0.0%	0.0%	2.6%	1.9%	0.0%	0.0%
Kono District	0.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.2%	0.6%	2.0%	1.8%
Moyamba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.7%	0.4%	0.1%	0.1%
Port Loko District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Pujehun District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	0.3%
Tonkolili District	0.0%	0.3%	0.0%	0.0%	0.0%	0.6%	0.0%	0.0%	1.5%	1.4%	0.1%	0.1%
Western Area Rural District	0.0%	0.0%	0.7%	1.5%	1.4%	0.5%	1.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Western Area Urban District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.6%	0.9%	1.1%	0.5%	1.5%	0.3%

2% or above

1% to 2%

Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100. The deviance is the difference in volume after removing the outlier. High levels of deviance can affect the plausibility of the data.

What you see: Heatmap showing how much service volume changed after replacing outliers with rolling averages.

Formula: % change = (adjusted - original) / original × 100

Interpretation: Values are typically negative (outliers removed reduce volume). Large adjustments warrant investigation into their source.

Completeness adjustment output

Deviance Due to Incompleteness

Percent change in volume due to completeness adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods - long acting	Family planning methods - short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	2.2%	4.7%	3.1%	4.7%	3.8%	2.2%	7.2%	9.0%	12.3%	18.4%	3.6%	8.3%
Bombali District	2.6%	3.3%	4.1%	10.4%	12.9%	7.4%	5.9%	5.6%	13.0%	5.6%	10.8%	8.6%
Bontho District	1.5%	5.2%	3.6%	8.3%	4.3%	1.6%	4.1%	4.0%	24.3%	8.3%	2.7%	6.1%
Falaba District	0.5%	2.8%	3.0%	6.4%	4.2%	0.9%	0.3%	0.3%	12.8%	7.5%	1.8%	4.7%
Kailahun District	0.5%	1.9%	1.3%	13.8%	11.9%	0.5%	0.4%	0.4%	8.3%	8.1%	2.8%	4.4%
Kambia District	0.8%	2.8%	0.4%	1.4%	1.3%	0.5%	0.6%	0.8%	9.6%	5.3%	0.3%	1.4%
Karene District	0.3%	0.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	8.3%	6.3%	2.3%	4.8%
Kenema District	0.3%	1.2%	0.7%	1.7%	1.3%	0.1%	0.2%	0.2%	5.6%	4.5%	0.5%	1.1%
Koinadugu District	2.3%	3.9%	2.7%	4.3%	2.6%	1.5%	2.1%	2.3%	41.1%	23.9%	13.1%	14.9%
Kono District	0.4%	2.1%	1.2%	5.6%	3.1%	0.6%	0.7%	0.9%	6.8%	7.3%	1.2%	6.5%
Moyamba District	1.4%	3.8%	2.9%	4.0%	3.3%	1.7%	2.0%	2.2%	13.9%	7.4%	2.8%	5.3%
Port Loko District	1.5%	2.0%	2.1%	2.9%	4.8%	1.6%	1.3%	1.4%	19.9%	9.8%	10.3%	8.8%
Pujehun District	0.3%	1.1%	2.1%	5.2%	4.0%	0.1%	0.2%	0.3%	7.0%	3.9%	0.3%	0.9%
Tonkolili District	1.2%	3.0%	4.6%	6.5%	2.7%	1.5%	1.8%	1.7%	9.6%	8.2%	2.5%	3.7%
Western Area Rural District	0.7%	0.8%	1.7%	1.7%	2.0%	0.6%	2.6%	2.4%	22.7%	10.1%	0.2%	0.6%
Western Area Urban District	6.2%	8.6%	5.7%	15.9%	13.5%	5.7%	6.6%	6.4%	19.9%	15.5%	4.5%	6.4%

3% or above

1% to 2%

Below 1%

Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. The deviance is the difference in volume after imputing incomplete data. High levels of deviance can affect the plausibility of the data.

What you see: Heatmap showing how much service volume changed after imputing missing data with rolling averages.

Formula: % change = (adjusted - original) / original × 100

Interpretation: Values are typically positive (imputation adds volume). Large adjustments indicate areas needing completeness improvement.

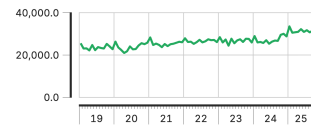
Service utilization analysis

Service utilization analysis tracks how many health services are being delivered over time, identifying trends, anomalies, and comparisons across areas.

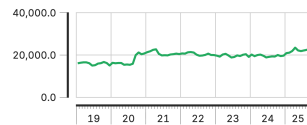
Service utilization over time

Jan 2019 to Sep 2025

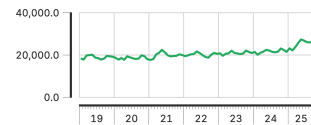
Antenatal client 1st visit



Antenatal client 4th visit



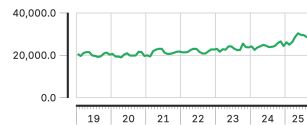
Institutional delivery (normal, assisted, and csection)



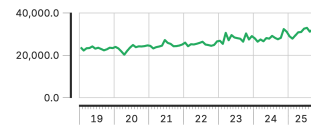
Postnatal care 1 (mothers)



BCG dose



Pentavalent 1st dose



Pentavalent 3rd dose



Outpatient visit



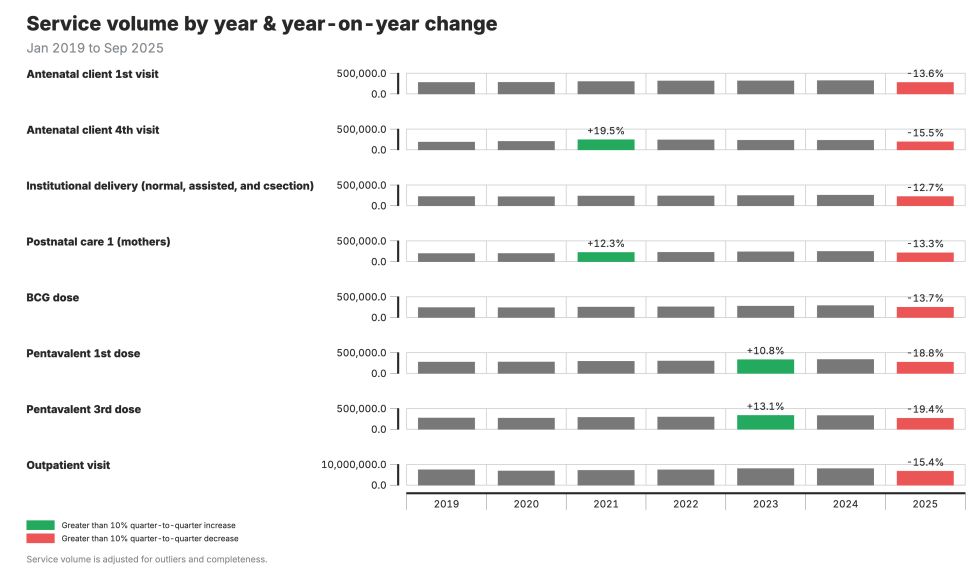
Number of services after both outlier and completeness adjustment
Service volume is adjusted for outliers and completeness.

What you see: Line chart showing absolute service volumes over time by indicator.

What it shows: Count of services delivered each month/quarter.

Interpretation: Look for overall trends (increasing/decreasing) and sudden drops or spikes that may need investigation.

Year-over-year change output



What you see: Heatmap comparing current period to same period last year, with changes >±10% flagged.

Formula: YoY change % = (this year - last year) / last year × 100

Interpretation: Flagged changes require follow-up - is this a real program change, data issue, or expected event?

Detecting service disruptions

Beyond year-over-year comparisons, we want to know: **Is service delivery on track, or has something disrupted it?**

The challenge: Raw service counts are hard to interpret. A drop in services could be a real disruption, or just normal seasonal variation. Different areas have different baseline volumes, making direct comparison difficult.

FASTR's solution: Use statistical modeling to estimate what service volume we would *expect* based on historical trends and seasonality, then compare actual volume to this expectation.

- **Disruption:** Observed volume significantly below expected
- **Surplus:** Observed volume significantly above expected

What it shows: Deviations from expected – disruptions (below) or surpluses (above).

Service coverage estimation

Coverage = services delivered ÷ target population

$$\text{Service coverage} = \frac{\text{Population who received the service}}{\text{Population who need the service (target population)}}$$

Numerator comes from DHIS2 data, count of services adjusted for outliers

Several options for estimating the denominator

HMIS tells us how many services were delivered (numerator), but not the target population size (denominator). Standard HMIS coverage uses catchment populations, which are often inaccurate. Surveys (DHS/MICS) provide reliable coverage but only every 3–5 years.

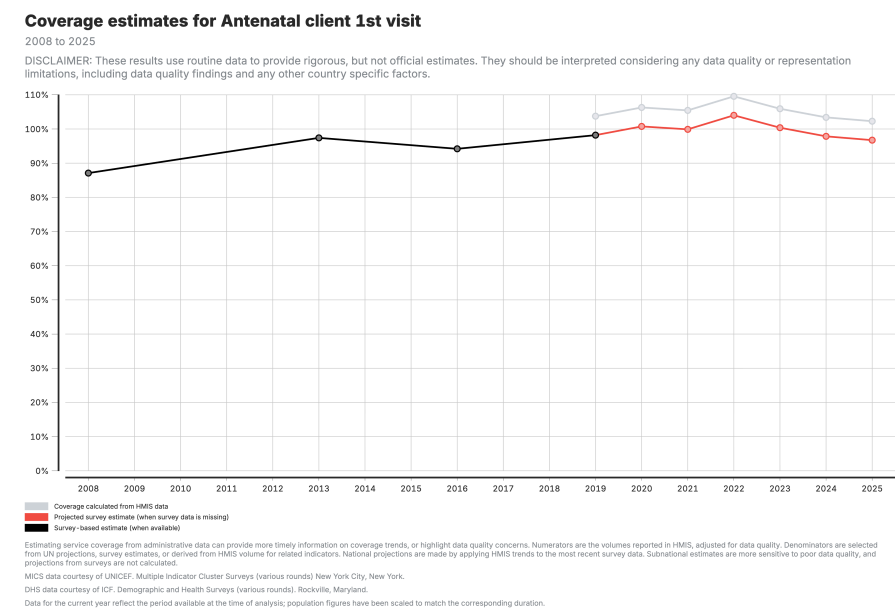
How FASTR estimates coverage

Calculate denominators multiple ways: From HMIS data, use service volumes combined with survey coverage to back-calculate target populations. For example, if 10,000 ANC1 visits and survey says 80% coverage, this implies ~12,500 pregnancies. Also calculate denominators from UN population projections using birth rates and demographic adjustments.

Validate against surveys: Calculate coverage using each denominator option, compare to survey benchmarks, and select the denominator with lowest error.

Project coverage forward: Anchor to the last survey value and apply year-over-year HMIS trends to extend estimates into post-survey years.

Coverage output: National trends



What you see: Line chart showing coverage over time. Black = survey, Grey = HMIS-derived, Red = projected.

Formula: Coverage % = (services / target population) × 100

Interpretation: Compare HMIS and survey values – large gaps suggest denominator issues. Projected values extend surveys using HMIS trends.



Session 7: Creating a project

Activity: Creating a Project



In this hands-on session, we will:

- Set up a new project
- Configure project settings
- Select indicators and time periods
- Apply best practices for project organization

Participants will create their first project

Session 8: Creating visualizations

Activity: Creating Visualizations



In this hands-on session, we will:

- Explore available chart types
- Create and customize visualizations
- Export charts for use in reports

Participants will build visualizations from their analysis



Lunch Break

90 minutes

Back at 14:00

Session 9: Creating reports

Activity: Creating Reports



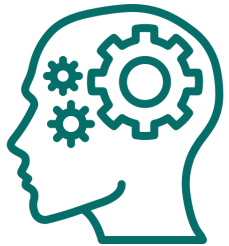
In this hands-on session, we will:

- Use report templates
- Generate automated reports
- Customize report content and layout

Participants will create their first quarterly report draft

Day 3

Recap of day 2



Agenda

Day 3 -- From Analysis to Action: Interpreting Results and Using FASTR for Decision-Making

Time	Agenda	Facilitator/Presenter
09:00–09:15	Recap and Day 3 agenda	GFF FASTR team
Session 10: Creating a Q4 2025 report		
09:15–12:30	Creating short and long reports with country context	GFF FASTR team
12:30–14:00	<i>Lunch Break</i>	
Session 10 (cont'd): Creating a Q4 2025 report		
14:00–15:30	Continue report creation with country context	GFF FASTR team
Session 11: Presenting reports		
15:30–16:30	Present reports, group feedback	GFF FASTR team

Session 10: Creating a Q4 2025 report

Generating quarterly reporting products

Content to be developed

This section will cover:

- Quarterly reporting workflow
- Using the FASTR platform for automated reports
- Quality assurance for reports
- Distribution and feedback mechanisms



Lunch Break

90 minutes

Back at 14:00

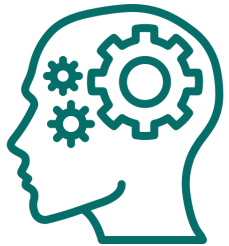
Session 11: Presenting reports

Presenting reports and group feedback

Content to be developed

Day 4

Recap of day 3



Agenda

Day 4 -- Designing the Health Facility Assessment

Time	Agenda	Facilitator/Presenter
09:00–09:15	Recap and Day 4 agenda	GFF FASTR team
Session 12: Overview of FASTR HFA phone survey		
09:15–10:15	HFA overview and questionnaire adaptation guidelines	GFF FASTR team
Session 13: Questionnaire adaptation to the Zambian context		
10:15–12:30	Review questionnaire + hands-on adaptation	GFF FASTR team
12:30–14:00	<i>Lunch Break</i>	
Session 13 (cont'd): Questionnaire adaptation		
14:00–15:00	Continue questionnaire adaptation (in groups)	GFF FASTR team
Session 14: Discussion: HFA adapted questionnaire		
14:30–15:30	Discuss adapted questionnaire in plenary	GFF FASTR team
Session 15: Discussion: HFA priorities and data use		
15:30–16:30	HFA priorities and data use case in Zambia	GFF FASTR team
Session 16: Action planning and wrap-up		
16:30–17:00	Key messages and wrap-up	GFF FASTR team

Session 12: Overview of FASTR HFA phone survey

Health Facility Assessment Phone Survey

Placeholder: Content to be developed



Lunch Break

90 minutes

Back at 14:00

Next Steps & Action Planning

Key actions to take after this working session:

- Finalize adapted phone survey questionnaire
- Complete first quarterly report draft
- Train remaining MoH team on data downloader tools
- Prepare roadmap for participation in Abuja workshop

Quarterly Reporting Cycle

Establish a sustainable reporting rhythm:

- Define quarterly report timeline and responsibilities
- Set up data extraction schedule
- Establish review and approval process
- Plan dissemination to stakeholders

Continued Platform Usage

Maintain momentum with the FASTR platform:

- Regular data updates and quality checks
- Expanding analysis to additional indicators
- Building capacity across MoH teams
- Documentation of lessons learned

Workshop Closing

Thank you for your participation!

Over the past four days, we have:

- Configured the FASTR platform for Zambia
- Created the first quarterly RMNCAH-N report
- Adapted the HFA questionnaire for local context
- Built capacity for ongoing analysis and reporting

Stay Connected

Continue the journey:

- Platform access and support
- Quarterly report submissions
- Multi-country workshop in Abuja
- Ongoing technical assistance from GFF team

Thank You

FASTR Working Session – Zambia
January 27–30, 2026
Lusaka

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